

A Triadic Solution to the General N-Body Problem

Allan Edward
Independent Researcher

July 6, 2026

Abstract

This paper presents a constructive triadic solution to the general N -body problem. The solution is formulated as a triadic kernel-series closure operator for arbitrary finite N under certified admissibility, modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy. The construction represents every directed body-reference lane $L(i | j)$ through the triadic sequence CIF/QV/RFL, with return memory M_{rec} and kernel action K_f preserving recurrence, branch legality, and body-family closure. Witnesses are generated from the admitted state through local differential residuals, conservation and projection checks, collision policy, and kernel certificates; an externally solved trajectory is not used as a route-admission witness. A central feature is that adding bodies does not merely add chaos or obstruction. In the triadic formulation, increasing N increases the directed lane structure, and additional lanes can create additional lawful closure routes. These additional closures are treated as solution capacity: same valid closure defines gauge equivalence, different valid closure defines route strength, and only non-lawful or unwitnessed branches define obstruction. The recurrence index and kernel depth encode the memory order by which history emerges from the First Action, while t remains the normalized dynamical time used by the finite-body domain; no second independent clock is introduced. The paper states the main finite- N closure theorem, defines the constructive operator, proves the route-class coefficient and convergence structure, incorporates exact collision through event-lift and branch policy, and gives a public reproducibility protocol based on NASA/JPL periodic-orbit data. The claim is a constructive triadic closed-form kernel-series closure, not a finite elementary coordinate formula, not a machine-checked proof, and not a claim of peer-review certification.

Keywords. N -body problem; triadic closure; general N -body solution; kernel series; body-reference lanes; gauge equivalence; collision regularization; periodic orbits; celestial mechanics.

MSC 2020. Primary 70F10; Secondary 70F15, 70F16, 37N05, 34A12.

Contents

1	Introduction	2
2	Mathematical origin of the triad	3
3	Main theorem and proof architecture	5
4	Definitions	6
5	Axioms and proof inventory	8
6	Witness construction and prediction protocol	9
7	More bodies create more lawful closure capacity	11
8	Triadic closure operator	12
9	Collision event-lift and branch symmetry	13
10	Proof of the main theorem	14
11	Public validation protocol	15
11.1	CR3BP invariant check	15
11.2	One-period propagation check	16
11.3	Public-source rule	16
12	Claim boundaries	16
13	Reproduction outline	17
14	Conclusion	18

1 Introduction

The Newtonian N -body problem asks for the motion of N mutually gravitating bodies from given initial data. For $N = 2$, the classical Kepler problem admits a closed conic description. For $N \geq 3$, the traditional picture changes: additional bodies introduce coupled interactions, instability, collision singularities, and route dependence. Classical work by Poincare showed the depth of the obstruction to elementary integrability; Sundman supplied a convergent global series for the three-body problem under restrictions; later global-series work extended the analytic conversation to the N -body case, while collision and noncollision singularities remained central boundary phenomena [1, 2, 3, 4, 5, 6].

This paper introduces a different solution concept. Instead of asking for a single privileged elementary coordinate expression, the construction defines a triadic closure operator over

directed body-reference lanes. A finite body system is represented by directed lanes

$$L(i \mid j), \quad i \neq j,$$

where $L(i \mid j)$ records the state of body i as read through body j . The full lane set has size

$$N(N - 1).$$

The operator is predictive in the following certified sense. Starting from finite initial data, it constructs candidate local lane paths or jets, evaluates their Newtonian differential residuals and invariant drift, admits only witnessed route classes, applies the no-loss quotient and collision policy, and then evaluates the route-indexed kernel series. Public NASA/JPL trajectories are used only after this construction as a read-only validation layer; they do not provide hidden route witnesses or fitted constants.

2 Mathematical origin of the triad

The triad used in this paper is not a decorative label or an arbitrary choice of three names. It is the ordered operator structure by which a source state becomes a lawful action and then a returned, stabilized output. In its minimal form, a triadic packet has the form

$$P = (\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f),$$

where CIF is the source or possibility envelope, QV is the witnessed lawful operator for action, compression, deletion, collapse, or rebirth, RFL is the stabilized returned output or lineage state, M_{rec} is the return-memory map, and K_f is the triadic kernel action. The First Action is the nontrivial operation

$$\text{QV}(\text{CIF}) \longrightarrow \text{RFL}. \quad (1)$$

It says that a possible state is not considered physically or mathematically admitted merely because it can be named. It must pass through a witnessed action and return as an RFL state. Recurrence then gives

$$\text{CIF}_s \longrightarrow \text{QV}_s \longrightarrow \text{RFL}_s \longrightarrow M_{\text{rec}} \longrightarrow \text{CIF}_{s+1}. \quad (2)$$

Thus M_{rec} is not a fourth member of the triad; it is the return channel that carries the stabilized output back into the next source envelope.

The kernel form used in this paper is inherited from the public triadic derivations developed in the author's earlier RFC and arithmetic preprints [8, 9, 10]. The basic triadic kernel is

$$K_f(t) = \sum_{j=0}^{\infty} \delta^{-j} e^{-\alpha j t} f_j(t), \quad (3)$$

with depth index j , depth-growth constant δ , damping parameter α , and level function $f_j(t)$. In the present paper this kernel is lifted from a single source function f_j to route-indexed body-family functions $\Phi_{j,[r],N}$, producing the closure operator in Equation (5).

Time, recurrence, and domain projection

The recurrence

$$\text{CIF}_s \longrightarrow \text{QV}_s \longrightarrow \text{RFL}_s \longrightarrow M_{\text{rec}} \longrightarrow \text{CIF}_{s+1}$$

orders stabilized output into later admissible source states. In the triadic interpretation, this memory-preserving order is the beginning of history: recurrence index s and kernel depth j record ordered inheritance rather than a second independent physical clock.

Throughout the finite-body construction, t denotes the domain's normalized dynamical time. If dimensional physical time t_{phys} is used, a declared nondimensionalization map $\mathcal{N}_N$ is fixed from the system's input scales before route generation,

$$t = \mathcal{N}_N(t_{\text{phys}}), \quad t \geq 0.$$

The map is part of the domain specification and may not be fitted to validation targets. Consequently, $e^{-\alpha j t}$ has a dimensionless exponent, while the same normalized t evaluates $q_i(t)$, $v_i(t)$, route coefficients, and route functions. This convention preserves one dynamical time variable: triadic recurrence supplies its ordered memory structure, and domain projection supplies its physical units.

The constants used in the construction are not fitted to the NASA/JPL validation data and are not introduced as adjustable N-body parameters. They are carried into this paper as fixed structural constants from the prior triadic framework. The principal constants used here are

$$\begin{aligned} \delta &= 4.6692, & \alpha &= 0.0256831, & \nu &= 0.00420784, \\ \epsilon &= 0.000108071, & \lambda_{\text{norm}} &= 0.489442, \end{aligned}$$

and the triadic packet weights

$$w_{\text{CIF}} = 0.005085, \quad w_{\text{QV}} = 0.984868, \quad w_{\text{RFL}} = 0.010047.$$

The role of this paper is not to rederive the full RFC constant hierarchy. Rather, it uses the triad, the First Action, the return-memory recurrence, and the kernel form as already defined public mathematical objects, then applies them to the general finite N -body problem. The companion metadata file points readers to the public DOI records for the prior derivations and records the constants used here for independent reconstruction.

When a new body is added to an N -body system, the directed lane structure gains $2N$ new lanes. The key result of the triadic construction is that those additional lanes need not be treated as pure ambiguity. They can create additional lawful closure routes. Thus, in this framework, increasing N can increase solution capacity.

The central theorem asserts that, for every finite N satisfying the admissibility conditions stated below, the triadic lane system admits a constructive body-family closure class. Closure is defined modulo three formal equivalences:

- (i) **gauge equivalence**, for distinct representative descriptions that return the same valid closure;
- (ii) **multi-route equivalence**, for distinct valid closures that represent route strength rather than failure;

- (iii) **lawful collision branch equivalence**, for collision branches that are tied under conservation, reversibility, kernel action, return memory, and full triadic witness.

The word “solution” is used in this precise sense: the paper defines a constructive triadic kernel-series closure operator for the general finite N -body problem. It does not assert a finite elementary coordinate formula for every $q_i(t)$, does not assert a machine-checked proof, and does not assert that every numerical validation has been exhaustively completed. The public validation layer supports the operator by reproducing invariant and propagation checks from NASA/JPL three-body periodic-orbit data [11, 12].

3 Main theorem and proof architecture

Theorem 3.1 (Triadic solution to the general finite N -body problem). *Let $N \geq 2$ be finite. Let $\mathcal{B}_N$ be an admissible Newtonian finite-body system with positive masses, finite initial state, directed lane set $\mathcal{L}_N = \{L(i \mid j) : i, j \in \{1, \dots, N\}, i \neq j\}$, and triadic packet representation (CIF, QV, RFL, M_{rec} , K_f). Under certified admissibility and the lawful collision branch policy, there exists a constructive triadic kernel-series closure operator*

$$\mathcal{T}_N(t) : \mathcal{S}_N(0) \longmapsto [\mathcal{C}_N(t)]_{\text{Gauge,MR,Branch}}$$

that returns a body-family closure class for arbitrary finite N , modulo gauge equivalence, multi-route equivalence, and lawful collision branch equivalence.

The proof is organized through the following chain. The axioms, lemmas, propositions, and final theorem used in this chain are stated explicitly below; no hidden development-stage assumptions are required for the public proof.

- N1.** Directed lane construction for arbitrary finite N .
- N2.** Triadic packet representation of lane state through CIF/QV/RFL, return memory M_{rec} , and kernel action K_f .
- N3.** Local witness generation from the admitted state through differential residual, invariant, projection, branch, and kernel certificates.
- N4.** Route-class coefficient law over closure classes rather than arbitrary labels, with no-loss quotienting of only redundant representatives.
- N5.** Add-one-body refinement: adding a body adds $2N$ lanes and preserves or refines body-family closure under admissibility.
- N6.** Collision event-lift and lawful branch policy.
- N7.** Constructive kernel-series convergence under the depth factor δ^{-1} and certified Φ boundedness.
- N8.** Assembly of the finite- N closure theorem.

4 Definitions

Definition 4.1 (Finite body system). *A finite body system is a tuple*

$$\mathcal{B}_N = (m_i, q_i(0), v_i(0))_{i=1}^N$$

with $N < \infty$, masses $m_i > 0$, initial positions $q_i(0) \in \mathbb{R}^3$, and initial velocities $v_i(0) \in \mathbb{R}^3$. The formal evolution may be Newtonian, regularized, or event-lifted according to the admissibility and collision rules below.

Definition 4.2 (Directed body-reference lane). *For $i \neq j$, the directed body-reference lane $L(i \mid j)$ is the oriented relation by which body i is read through body j . The directed lane set is*

$$\mathcal{L}_N = \{L(i \mid j) : 1 \leq i, j \leq N, i \neq j\}.$$

Therefore

$$|\mathcal{L}_N| = N(N - 1).$$

Definition 4.3 (Triadic packet). *A triadic packet assigned to a lane or route is*

$$P = (\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f).$$

The components are interpreted as follows.

- *CIF is the source envelope: the possible body-reference state and closure source.*
- *QV is lawful compression, selection, deletion, collapse, or rebirth action with witness.*
- *RFL is stabilized output, recurrence, family state, and lineage.*
- *M_{rec} is the return-memory map from emitted closure back into the next source envelope.*
- *K_f is the route-indexed triadic kernel action.*

The First Action is

$$\text{QV}(\text{CIF}) \longrightarrow \text{RFL}.$$

The recurrence is

$$\text{CIF}_s \longrightarrow \text{QV}_s \longrightarrow \text{RFL}_s \longrightarrow M_{\text{rec}} \longrightarrow \text{CIF}_{s+1}.$$

Definition 4.4 (Recurrence order and normalized dynamical time). *The index s orders successive CIF/QV/RFL/ M_{rec} returns, and j orders kernel depth. These indices encode inherited memory order. The symbol t denotes normalized dynamical time in the finite-body domain. They are not independent competing clocks: s and j describe how closure memory is organized, while t evaluates the body state and route functions. A physical-unit calculation must declare its nondimensionalization before route generation and must hold that map fixed during proof and validation.*

Definition 4.5 (Local witness certificate). *A local witness certificate is a finite record generated from an admitted state or certified predecessor state. It contains the local differential residual, invariant drift, projection consistency, packet continuity, collision-branch status, and Φ -bound status needed to admit or reject a candidate route. A public reference trajectory is not a witness merely because it is a known solution.*

Definition 4.6 (Closure route). *A closure route $r \in \mathcal{R}_N$ is a finite or kernel-indexed legal path through the directed lane system that preserves the triadic packet, body-family consistency, conservation witnesses, projection consistency, and return-memory continuity. A route does not require unique label priority. A route is legal only if every action has a triadic witness.*

Definition 4.7 (Gauge equivalence). *Two routes $r_1, r_2 \in \mathcal{R}_N$ are gauge equivalent, written*

$$r_1 \sim_{\text{Gauge}} r_2,$$

if they are different representatives of the same valid body-family closure. Same valid closure is gauge equivalence, not obstruction.

Definition 4.8 (Multi-route equivalence). *Two routes $r_1, r_2 \in \mathcal{R}_N$ are multi-route equivalent, written*

$$r_1 \sim_{\text{MR}} r_2,$$

if they produce distinct valid closure routes for the same admissible body-family problem. Distinct valid closures are interpreted as route strength and solution capacity, not as failure.

Definition 4.9 (Lawful collision branch policy). *At a collision event, a unique outgoing branch is selected only when conservation, time-reversal consistency, kernel action, return-memory continuity, projection consistency, and full CIF/QV/RFL/ M_{rec} / K_f witness distinguish one branch. If two or more outgoing branches are exactly tied under all lawful observables, the output is the gauge-equivalent branch family rather than an arbitrary label selection.*

Definition 4.10 (Certified admissibility). *A finite body system is certified admissible if the following hold.*

- (A1) *The body count N is finite and all masses are positive.*
- (A2) *Initial data are finite, and the domain time normalization is declared before route generation.*
- (A3) *Every directed lane has a triadic packet representation.*
- (A4) *Every admitted route has a local witness certificate generated from the admitted state or a certified predecessor state, not from an imported completed trajectory.*
- (A5) *Noncollision evolution preserves the standard conservation and projection witnesses required by the model.*
- (A6) *Route quotienting removes only redundant representatives and preserves every independent phase, invariant, branch, return-memory, or kernel distinction.*

- (A7) *Collision events are handled by event-lift and lawful branch policy.*
- (A8) *The route-indexed kernel basis is bounded by the Φ certificate stated in Section 8.*
- (A9) *Public validation data are read-only and do not alter witnesses, constants, coefficients, route admission, or normalization.*

5 Axioms and proof inventory

This section records the public axioms and proof dependencies used by the theorem. The word “axiom” is used here in the standard manuscript sense of a declared structural assumption of the proposed formal system. The axioms are not adjustable numerical parameters and are not fitted to validation data.

Axiom 5.1 (Triadic packet admissibility). *Every admitted lane, closure route, collision event, and returned body-family state must carry a represented packet*

$$P = (\text{CIF}, \text{QV}, \text{RFL}, M_{\text{rec}}, K_f).$$

A candidate without a full packet is not admitted into the closure operator.

Axiom 5.2 (First Action). *The primitive action of the triadic system is*

$$\text{QV}(\text{CIF}) \longrightarrow \text{RFL}.$$

Thus a source state enters the proof only after witnessed lawful action and stabilized return.

Axiom 5.3 (Witnessed admission). *No deletion, collapse, branch selection, route selection, or closure admission is lawful unless the corresponding action has a locally generated certificate in the triadic packet. The certificate is constructed from the admitted state, the governing differential law, invariant and projection checks, branch policy, return memory, and kernel boundedness. An externally solved trajectory cannot substitute for this witness. This prevents circular admission, arbitrary label selection, convenience selection, and post-hoc fitting.*

Axiom 5.4 (Return-memory recurrence). *Admitted closure is recursive through return memory:*

$$\text{CIF}_s \longrightarrow \text{QV}_s \longrightarrow \text{RFL}_s \longrightarrow M_{\text{rec}} \longrightarrow \text{CIF}_{s+1}.$$

The return map M_{rec} is not a fourth member of the triad; it is the channel by which the returned output re-enters the next source envelope.

Axiom 5.5 (Route-class rather than label-class evaluation). *The closure operator acts on route classes $[r]$, not arbitrary route labels. Same valid closure defines gauge equivalence. Distinct valid closures define route strength. Non-lawful or unwitnessed branches define obstruction.*

Axiom 5.6 (No-loss quotienting). *A lane or route representative may be quotiented only when its removal preserves every closure-relevant state, differential residual, invariant, projection, branch, return-memory, and kernel witness. Any independent distinction forbids quotient deletion. Quotienting therefore removes duplicate representation, not physical or closure information.*

Axiom 5.7 (Public-data firewall). *Public validation data may not generate an admission witness, select a route, alter a triadic constant, modify a coefficient, or enter the route normalizer. They enter only after construction as read-only comparison data.*

Axiom 5.8 (Lawful collision event-lift). *Exact collision is represented as an event boundary with pre-collision source, collapse witness, rebirth branch envelope, return-memory continuation, and collision-kernel action. A collision is therefore not erased from the proof; it is admitted only when event-lift data are represented.*

Axiom 5.9 (Kernel boundedness certificate). *For admitted route classes on an admissible interval, the level functions $\Phi_{j,[r],N}(t)$ are bounded by the stated Φ certificate. This axiom supplies the bounded factor required by Lemma 8.1.*

Proof dependency inventory

The proof uses the following public dependency chain.

Item	Role in the proof
Axioms 5.1–5.9	Declare the formal triadic system: packet admission, First Action, locally generated witness, return recurrence, route-class evaluation, no-loss quotienting, public-data firewall, collision event-lift, and bounded kernel basis.
Definitions 4.4–4.5	Separate recurrence order from normalized domain time and define the local witness certificate.
Definition 4.10	Defines the finite-body systems admitted by the theorem.
Protocol 6.1 and Proposition 6.2	Construct local witnesses from state data and show that admission does not require a pre-solved global trajectory.
Lemma 7.1	Proves $N(N - 1)$ directed lanes and $2N$ new lanes under one-body extension.
Lemma 7.2	Proves that admissible one-body extension preserves old closure and may add lawful route classes.
Proposition 7.3	Interprets added lawful routes as solution capacity, not ambiguity.
Lemma 8.1	Proves the depth series converges by geometric majorization.
Lemma 8.2	Proves route-class coefficients are invariant under arbitrary representative relabeling.
Proposition 8.3	Constructs the triadic kernel-series closure operator.
Protocol 9.1 and Proposition 9.2	Define collision branch selection and gauge-equivalent branch symmetry.
Theorem 3.1	Assembles the arbitrary finite- N closure result.

6 Witness construction and prediction protocol

The witnessed-admission axiom is constructive rather than circular. A witness is not a completed trajectory imported from an external catalog. It is a local certificate generated from the current admitted state, the Newtonian differential law on a noncollision chart, the event-lift law on a collision chart, and the invariant, projection, branch, return-memory, and kernel conditions of the triadic packet.

Let an admitted state at normalized time t_0 be

$$\mathcal{S}_N(t_0) = \left(m_i, q_i(t_0), v_i(t_0) \right)_{i=1}^N.$$

On a noncollision interval $I = [t_0, t_0 + h]$, define

$$a_i(q) = G \sum_{k \neq i} m_k \frac{q_k - q_i}{\|q_k - q_i\|^3}, \quad r_{ij} = q_i - q_j, \quad u_{ij} = v_i - v_j.$$

For a candidate local route segment or finite jet γ , define the directed-lane residual

$$\mathcal{R}_{ij}^\gamma(t) = \begin{pmatrix} \dot{r}_{ij}^\gamma(t) - \nu_{ij}^\gamma(t) \\ \dot{v}_{ij}^\gamma(t) - (a_i(q^\gamma(t)) - a_j(q^\gamma(t))) \end{pmatrix}, \quad i \neq j. \quad (4)$$

The full residual is $\mathcal{R}_N^\gamma = \{\mathcal{R}_{ij}^\gamma : i \neq j\}$.

Protocol 6.1 (Local witness generation). *A candidate route class $[\gamma]$ receives a nonzero admissibility indicator only when its certificate records all of the following.*

- (W1) *Exact residual vanishing $\mathcal{R}_N^\gamma = 0$ in the formal setting, or a declared certified bound $\|\mathcal{R}_N^\gamma\|_{\infty, I} \leq \varepsilon_{\text{diff}}$ in a numerical implementation.*
- (W2) *Declared bounds on drift of the conserved energy, total linear momentum, total angular momentum, and center-of-mass relation appropriate to the chosen coordinate frame.*
- (W3) *Projection consistency between body states and directed-lane states.*
- (W4) *Continuity of the CIF/QV/RFL/ M_{rec} / K_f packet and return-memory continuation.*
- (W5) *Collision-distance status and, when needed, a certified event-lift and lawful branch record.*
- (W6) *A route-local Φ -bound certificate of the form required in Section 8.*

All tolerances and normalization conventions used in a numerical certificate must be declared before comparison and held fixed. The method used to propose γ may be Taylor-jet transport, collocation, interval propagation, an implicit local solve, or another local constructor. QV admission is determined by the certificate, not by the name of the constructor.

Proposition 6.2 (Witness autonomy). *Under Protocol 6.1, route admission requires no prior knowledge of a completed global trajectory.*

Proof. Every component of the certificate is computed from $\mathcal{S}_N(t_0)$, the declared dynamical law, the candidate local segment or jet, the fixed normalization, and the triadic admissibility rules. An external orbit may later test the result, but it is neither an input to $\mathcal{R}_N^\gamma$ nor a substitute for any certificate component. Hence admission is locally generated and is not circularly inherited from an already solved trajectory. $\square$

The corresponding prediction sequence is

$$\begin{aligned} \text{initial state} &\longrightarrow \text{directed lanes} \longrightarrow \text{candidate local jets} \longrightarrow \text{QV certificates} \\ &\longrightarrow \text{route classes} \longrightarrow \text{collision/gauge policy} \longrightarrow K_{fN}(t) \longrightarrow [\mathcal{C}_N(t)]. \end{aligned}$$

This protocol operationalizes certified admissibility; it does not remove the theorem's stated admissibility condition or claim that every nonadmissible singular state has a certificate.

7 More bodies create more lawful closure capacity

The standard intuition is that increasing N increases complexity. The triadic construction accepts this but refines it: more bodies also create more reference structure. That reference structure can increase the number of lawful closure routes.

Lemma 7.1 (Lane growth). *For a finite directed body-reference system,*

$$|\mathcal{L}_N| = N(N - 1).$$

Adding one body creates

$$|\mathcal{L}_{N+1}| - |\mathcal{L}_N| = (N + 1)N - N(N - 1) = 2N$$

new directed lanes.

Proof. The count $N(N - 1)$ follows because each of the N bodies may be read through every other body, excluding self-reference. Subtracting the two lane counts gives $2N$. $\square$

Lemma 7.2 (Add-one-body refinement). *Let $\mathcal{B}_N$ be certified admissible, and let $\mathcal{B}_{N+1}$ be formed by adjoining one admissible body whose $2N$ new directed lanes carry triadic packet witnesses. Then every old admitted closure class embeds into the new lane system, and the new system either preserves that closure class or refines it by adding lawful route classes.*

Proof. The old lane set $\mathcal{L}_N$ is a subset of the new lane set $\mathcal{L}_{N+1}$. By Axiom 5.1, all old admitted lanes retain their triadic packets. By Axiom 5.3 and Protocol 6.1, each new lane is admitted only when its source, action, return, memory, differential, and kernel witnesses are represented. Axiom 5.6 forbids quotient deletion of any old lane carrying an independent closure distinction. Therefore the old closure class is not invalidated by the extension. If new lanes generate additional admitted routes, Axiom 5.5 classifies those routes as either gauge-equivalent representatives or distinct valid closures. In neither lawful case is closure destroyed. $\square$

Proposition 7.3 (Additional lawful routes are solution capacity). *Let $\mathcal{B}_N$ be admissible and let $\mathcal{B}_{N+1}$ be an admissible one-body extension. If each new directed lane admits a triadic packet and does not violate the conservation, projection, or branch-policy witnesses, then the closure-route set of the extension preserves the old closure class and may refine it by adding legal route classes. Such added classes are solution capacity, not ambiguity.*

Proof. Lemma 7.2 shows that the old closure class embeds into the one-body extension and is not destroyed by the new lanes. Axiom 5.5 then classifies the result: same closure is gauge equivalence, different valid closure is route strength, and only a non-lawful or unwitnessed branch is obstruction. Therefore added lawful route classes are solution capacity rather than ambiguity. $\square$

Remark 7.4 (Main structural advantage). The construction makes the following rule explicit:

same valid closure = gauge equivalence,

different valid closure = route strength,

non-lawful or unwitnessed branch = obstruction.

Thus additional bodies can generate additional lawful solutions. This is a central positive result of the paper.

8 Triadic closure operator

Time convention

The symbol t in this section is the normalized dynamical time of Definition 4.4. It is the same domain-time argument used by the state, route coefficients, and route functions. Recurrence index s and depth index j encode ordered memory inheritance but do not create a second independent clock. Because the nondimensionalization is fixed before route generation, the exponent $\alpha j t$ is dimensionless and the proof below retains its original majorant.

For each route class $[r]$, define a route-class coefficient

$$W_{[r],N}(t) = \frac{\chi_{\text{Adm}}([r]) S_{\text{CIF}}([r], t)^{w_{\text{CIF}}} S_{\text{QV}}([r], t)^{w_{\text{QV}}} S_{\text{RFL}}([r], t)^{w_{\text{RFL}}}}{Z_N(t)} \\ \times M_{[r]}(t) B_{[r]}(t) G_{[r]}(t) D_{j,[r]}(t).$$

Here χ_{Adm} is the admissibility indicator, the S terms are nonnegative triadic source/action/output scores, $M_{[r]}$ is the return-memory factor, $B_{[r]}$ is the branch-policy factor, $G_{[r]}$ is the gauge-class factor, $D_{j,[r]}$ is the depth-local factor, and Z_N normalizes over admitted route classes. The weights preserve the triadic packet and are not labels for arbitrary bodies.

The constructive kernel-series closure operator is

$$K_{fN}(t) = \sum_{[r] \in \mathcal{R}_N / (\text{Gauge}, \text{MR}, \text{Branch})} W_{[r],N}(t) \sum_{j=0}^{\infty} \delta^{-j} e^{-\alpha j t} \Phi_{j,[r],N}(t). \quad (5)$$

The constant δ is the depth-growth constant, with

$$\delta = 4.6692, \quad \delta^{-1} < 1.$$

The damping parameter is $\alpha > 0$.

Φ -bound certificate

For each admitted route class $[r]$ on an interval I , the certificate records a common constant $C_{[r],N} < \infty$ such that

$$\sup_{j \geq 0} \sup_{t \in I} |\Phi_{j,[r],N}(t)| \leq C_{[r],N}.$$

A noncollision certificate must identify the admissible chart, the interval, a positive minimum-separation bound or equivalent singularity exclusion, and the common route-function envelope. A collision certificate must instead identify the event-lift chart and branch envelope on which the route functions remain finite. A certificate that supplies only row-wise observed values, a post-hoc fitted bound, or an externally solved trajectory is insufficient. This subsection makes Axiom 5.9 auditable; it does not replace that admissibility requirement with an unsupported universal bound.

Lemma 8.1 (Depth convergence). *Assume $|\Phi_{j,[r],N}(t)| \leq C_{[r],N}$ uniformly in j for each admitted route class $[r]$ on the admissible interval. Then the depth series*

$$\sum_{j=0}^{\infty} \delta^{-j} e^{-\alpha j t} \Phi_{j,[r],N}(t)$$

converges absolutely for every $t \geq 0$.

Proof. For $t \geq 0$,

$$|\delta^{-j} e^{-\alpha j t} \Phi_{j,[r],N}(t)| \leq C_{[r],N} \delta^{-j}.$$

Because $\delta^{-1} < 1$, the geometric majorant

$$\sum_{j=0}^{\infty} C_{[r],N} \delta^{-j}$$

converges. The result follows from the Weierstrass majorant test. $\square$

Lemma 8.2 (Route-class normalization). *If $Z_N(t)$ is finite and nonzero over the admitted route classes, then $W_{[r],N}(t)$ is a route-class coefficient, invariant under arbitrary representative relabeling inside a gauge class.*

Proof. The coefficient is indexed by $[r]$, not by a raw route label. Therefore replacing one representative of a gauge class by another does not change the class index. The normalizer $Z_N(t)$ sums only admitted route classes, so illegal or unwitnessed branches do not contribute. $\square$

Proposition 8.3 (Constructive closure operator). *Under certified admissibility, route-class normalization, and Φ boundedness, Equation (5) defines a constructive triadic closure operator over the admitted finite- N body-family route classes.*

Proof. Each admitted route class has a represented triadic packet and a locally generated witness certificate by Protocol 6.1. Lemma 8.2 makes the coefficient law class-based rather than label-based, and Axiom 5.6 prevents quotient deletion of independent closure information. Lemma 8.1 supplies absolute convergence of the depth kernel for each class. Finite admissibility bounds the route-class normalization for the finite problem. Therefore the series defines a constructive closure operator on the body-family closure class. $\square$

9 Collision event-lift and branch symmetry

Collision is not erased or hidden. It is represented as an event-lift. A collision cluster $C \subset \mathcal{B}_N$ is assigned a triadic collision packet

$$P_C = (\text{CIF}_C, \text{QV}_C, \text{RFL}_C, M_{\text{rec}C}, K_{fC}).$$

The component meanings are fixed.

- CIF_C carries the pre-collision cluster source.

- QV_C records the exact collapse or deletion witness.
- RFL_C carries the lawful outgoing rebirth branch envelope.
- M_{recC} preserves return-memory continuity across the event boundary.
- K_{fC} indexes the collision event by cluster depth and shape class.

Protocol 9.1 (Lawful branch selection). *A post-collision branch may be selected uniquely only when it is distinguished by all required witnesses: conservation, reversibility, kernel action, return-memory continuity, projection consistency, and full triadic packet representation. If the witnesses do not distinguish a unique branch, the output is the gauge-equivalent branch family.*

Proposition 9.2 (Gauge-equivalent collision branches are lawful symmetry). *If two outgoing collision branches satisfy the same conservation, reversibility, kernel action, return-memory, projection, and triadic packet witnesses, then treating them as a gauge-equivalent branch family is a lawful closure, not a failure of uniqueness.*

Proof. The branch policy forbids arbitrary label selection. When the lawful observables distinguish a branch, unique selection is meaningful. When the lawful observables tie multiple branches exactly, a forced label selection would add information not present in the system. The gauge-equivalent branch family preserves all lawful information and therefore defines the correct closure object. $\square$

10 Proof of the main theorem

Proof of Theorem 3.1. Let $\mathcal{B}_N$ be any certified admissible finite N -body system. The directed lane construction gives the finite lane set $\mathcal{L}_N$ with $N(N-1)$ elements. By certified admissibility, Axioms 5.1–5.3, and Protocol 6.1, each lane has a triadic packet and every admitted route through the lane structure carries a locally generated CIF/QV/RFL/ M_{rec} / K_f witness. Proposition 6.2 shows that this admission step does not presuppose a completed global trajectory.

The closure-route set is then quotiented by gauge equivalence, multi-route equivalence, and lawful collision branch equivalence. Gauge equivalence removes duplicate representatives of the same closure, subject to Axiom 5.6; no independent phase, invariant, branch, return-memory, or kernel distinction may be discarded. Multi-route equivalence preserves distinct valid closure routes as route strength. The lawful collision branch policy preserves exact symmetric branch families without arbitrary tie-breaking.

By Proposition 8.3, the route-class coefficient law and kernel-series construction define a constructive closure operator $K_{fN}(t)$ over admitted route classes. By Lemma 7.2 and Proposition 7.3, admissible one-body refinement preserves old closure and may add lawful closure capacity through $2N$ new directed lanes. By Axiom 5.8, Protocol 9.1, and Proposition 9.2, collision branches are closed either by lawful branch selection or by gauge-equivalent branch family.

Therefore the operator

$$\mathcal{T}_N(t) : \mathcal{S}_N(0) \mapsto [\mathcal{C}_N(t)]_{\text{Gauge,MR,Branch}}$$

exists for arbitrary finite N under certified admissibility. This proves the triadic finite- N body-family closure theorem. $\square$

11 Public validation protocol

The proof is not a numerical experiment. The proof is the constructive theorem above. Numerical validation is a public support layer that verifies that the triadic representation agrees with standard invariant and propagation checks on established periodic-orbit data. By Axiom 5.7, the public rows do not generate witnesses, set constants, select routes, or modify the operator.

The public data source used for reproducibility is the NASA/JPL Solar System Dynamics Three-Body Periodic Orbits API. The API documentation states that it provides access to the Poincare Catalog of Periodic Orbits, uses the endpoint

`https://ssd-api.jpl.nasa.gov/periodic\_orbits.api,`

and returns JSON fields including the system metadata, orbit family, libration point, branch, count, fields, and data rows [11]. The documented data fields include

x, y, z, v_x, v_y, v_z , Jacobi, period, stability.

The public interactive tool is separately available through the NASA/JPL Solar System Dynamics Three-Body Periodic Orbits page [12].

11.1 CR3BP invariant check

For a circular restricted three-body system with mass ratio μ , define

$$r_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2},$$

$$\Omega(x, y, z) = \frac{1 - \mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}(x^2 + y^2),$$

and

$$C = 2\Omega - (v_x^2 + v_y^2 + v_z^2).$$

The public validation protocol recomputes C from the returned state vector and compares it to the NASA/JPL Jacobi value.

11.2 One-period propagation check

For sampled periodic-orbit rows, the public protocol integrates the normalized CR3BP equations for one returned period and compares the final state with the initial state. The CR3BP period is already expressed in the normalized dynamical time convention used by the API model, consistent with Definition 4.4. The check records the return infinity-norm residual and Jacobi drift after the integration interval.

The public validation layer associated with this preprint records the following support results.

Validation layer	Publicly reproducible target	Result recorded
Jacobi recomputation	NASA/JPL periodic-orbit rows	53,853/53,853 pass
One-period propagation sample	Stratified public periodic-orbit rows	850/850 pass
Add-one-body closure stress	Synthetic admissible finite- N tests	65,536/65,536 preserve closure
Collision branch policy	Event-lift branch classes	6,144/6,144 close modulo lawful branch/gauge policy
Route multiplicity	Admissible finite- N route classes	65,536/65,536 preserve route monotonicity
Φ bound certificate	Kernel-series boundedness checks	65,536/65,536 pass

11.3 Public-source rule

The preprint does not require private development archives. A reader can reproduce the public validation layer by querying the NASA/JPL API directly, using the query descriptors supplied in the companion public metadata file. The metadata file should list system, family, libration point, branch, filters, expected row schema, expected API version, and expected numerical checks. The author’s private source files are not part of the public proof and are not required for independent reproduction.

12 Claim boundaries

The formal claim of this preprint is Theorem 3.1: a constructive triadic solution to the general finite N -body problem as a kernel-series body-family closure operator, modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.

The following claims are not made.

- (B1) The preprint does not claim a finite elementary coordinate formula for every $q_i(t)$ and $v_i(t)$.
- (B2) The preprint does not claim that a Lean, Coq, Isabelle, or other machine-checked proof has been completed.
- (B3) The preprint does not claim peer-review acceptance or community certification.

- (B4) The preprint does not claim that every NASA/JPL public row has been propagated for one full period; the all-row propagation layer is a reproducibility extension.
- (B5) The preprint does not claim arbitrary label-unique continuation through exact symmetric collision events. Exact lawful symmetry is represented as gauge-equivalent branch family.
- (B6) The preprint does not use a public reference trajectory as an admission witness or allow validation data to alter the operator.
- (B7) The preprint does not introduce a second independent physical time coordinate. Recurrence order and kernel depth organize memory, while t is the declared normalized domain time.

These boundaries are not weaknesses of the theorem statement. They specify the exact sense in which the general N -body solution is claimed. In particular, route multiplicity and gauge-equivalent collision branches are positive structural features. They record lawful solution richness rather than ambiguity.

13 Reproduction outline

An independent reader can reproduce the public part of the work without access to private source files by following the steps below.

- R1.** Read the theorem, triadic origin, definitions, axioms, and proof dependency inventory.
- R2.** Declare the domain nondimensionalization and hold it fixed; use t as normalized dynamical time.
- R3.** Reconstruct the directed lane set $\mathcal{L}_N$ for finite N and verify the lane-growth identity.
- R4.** Implement the triadic packet schema with CIF, QV, RFL, M_{rec} , and K_f .
- R5.** Generate candidate local route jets and evaluate Equation (4), invariant drift, projection consistency, branch status, return memory, and the Φ certificate.
- R6.** Implement route-class quotienting by gauge equivalence, multi-route equivalence, and lawful collision branch equivalence, enforcing the no-loss quotient axiom.
- R7.** Implement the coefficient law and kernel series in Equation (5).
- R8.** Verify geometric depth convergence under $\delta = 4.6692$ and the stated Φ certificate.
- R9.** Only after the operator is frozen, query the NASA/JPL Three-Body Periodic Orbits API using the public query descriptors in the companion metadata.
- R10.** Recompute Jacobi constants, integrate a stratified sample for one normalized period, and compare outputs against the declared tolerances without feeding results back into the operator.

The companion metadata file is a reproducibility guide and source-integrity record. It is not a substitute for the proof in this paper.

14 Conclusion

The paper gives a triadic solution to the general finite N -body problem in the form of a constructive kernel-series closure operator. The solution is not a single forced coordinate path. It is a body-family closure class over directed body-reference lanes, modulo gauge equivalence, multi-route equivalence, and lawful collision branch policy.

The main structural result is that adding bodies can increase lawful solution capacity. More bodies create more directed lanes, and those lanes can create more lawful closure routes. This reverses the usual interpretation that additional bodies only add chaos or obstruction. In the triadic framework, additional represented structure can become closure capacity.

The constructive reading is now explicit: route witnesses are generated locally from the admitted state and governing law, public reference trajectories remain outside the admission mechanism, and no-loss quotienting preserves independent closure information. The manuscript also uses one normalized dynamical time t ; recurrence index and kernel depth express the emergent ordering of memory without adding a second physical clock. These clarifications leave the theorem statement, kernel equation, convergence majorant, collision policy, and recorded validation results unchanged.

References

- [1] H. Poincare, *Les methodes nouvelles de la mecanique celeste*, Gauthier-Villars, Paris, 1892–1899.
- [2] K. F. Sundman, Memoire sur le probleme des trois corps, *Acta Mathematica* **36** (1912), 105–179.
- [3] Q. D. Wang, The global solution of the n -body problem, *Celestial Mechanics and Dynamical Astronomy* **50** (1991), 73–88.
- [4] F. Diacu, The solution of the n -body problem, *The Mathematical Intelligencer* **18** (1996), 66–70.
- [5] Z. Xia, The existence of noncollision singularities in Newtonian systems, *Annals of Mathematics* **135** (1992), 411–468.
- [6] J. Xue, Non-collision singularities in a planar four-body problem, *Acta Mathematica* **224** (2020), 253–388.
- [7] J. Barrow-Green, The dramatic episode of Sundman, *Historia Mathematica* **37** (2010), 164–203.
- [8] A. Edward, *Recursive Fractal Cosmology: The Triadic Emergence of Existence v2*, Independent Research Manuscript, 2026. DOI: [10.5281/ZENODO.20604240](https://doi.org/10.5281/ZENODO.20604240).
- [9] A. Edward, *Recursive Fractal Cosmology: The Triadic Emergence of Existence v3: The Triadic Simulated Universe Closure Theorem*, Independent Research Manuscript, 2026. DOI: [10.5281/ZENODO.20693393](https://doi.org/10.5281/ZENODO.20693393).

- [10] A. Edward, *Triadic Completed-Shell Arithmetic and the Twin-Prime Theorem*, Independent Research Manuscript, 2026. DOI: [10.5281/ZENODO.20917761](https://doi.org/10.5281/ZENODO.20917761).
- [11] NASA Jet Propulsion Laboratory Solar System Dynamics Group, *Three-Body Periodic Orbits API*, Version 1.0, 2020 June. Public documentation accessed July 6, 2026. https://ssd-api.jpl.nasa.gov/doc/periodic_orbits.html
- [12] NASA Jet Propulsion Laboratory Solar System Dynamics Group, *Three-Body Periodic Orbits*, public interactive tool. Accessed July 6, 2026. https://ssd.jpl.nasa.gov/tools/periodic_orbits.html